MILITARY SPECIFICATION SHEET

SWITCHES, PUSHBUTTON, ILLUMINATED, 2 AMPERES, 2PDT, 4PDT AND INDICATOR, 4 LAMP, SUNLIGHT READABLE (EMI/RFI SHEILDING, MODULAR CONSTRUCTION)

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-22885.

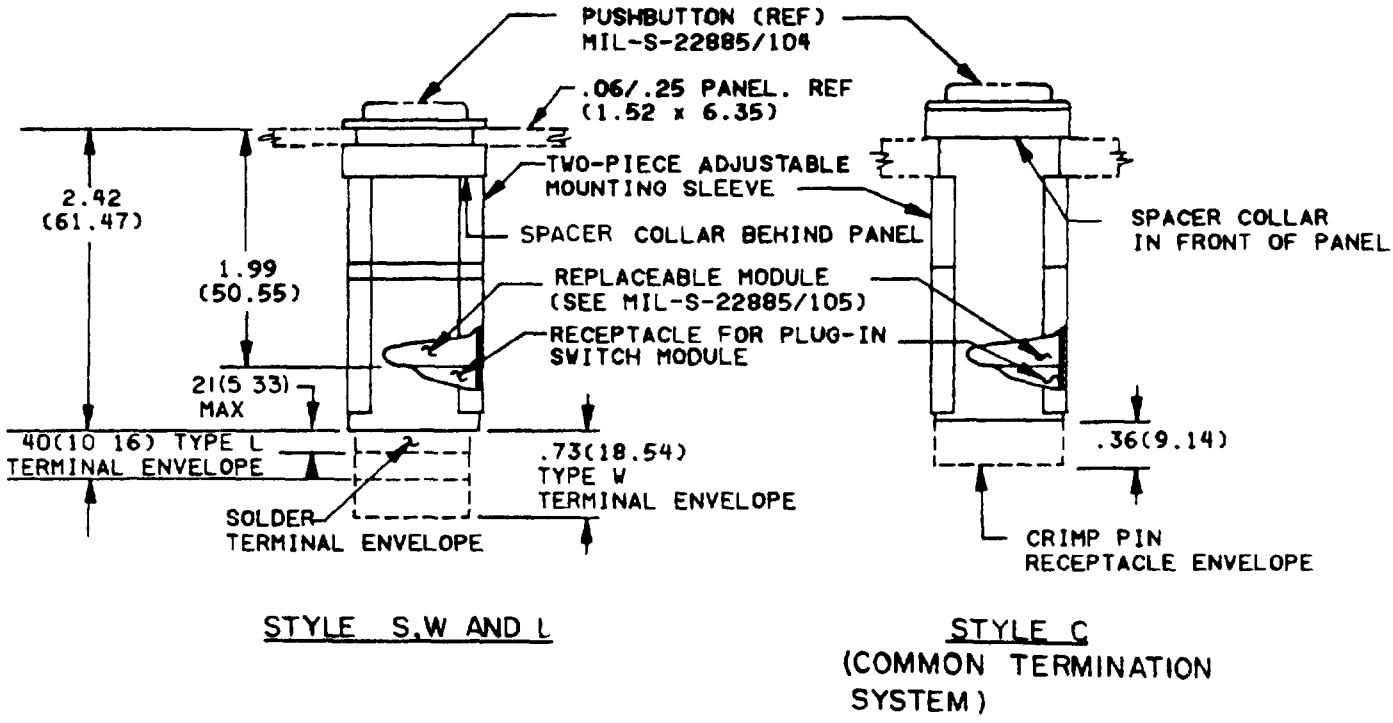
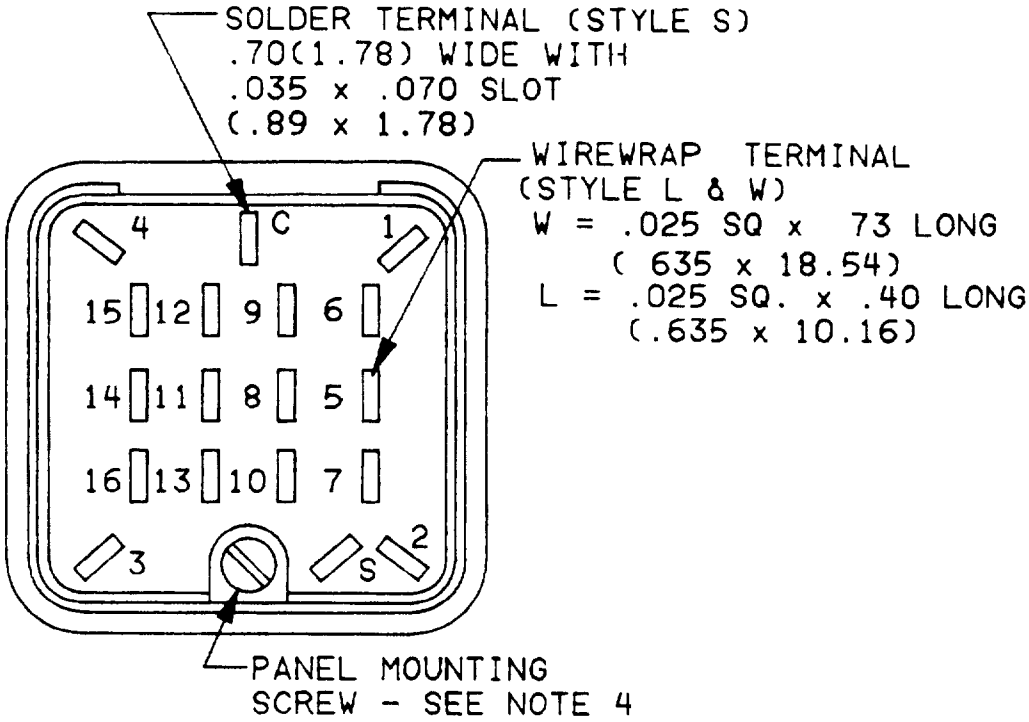


FIGURE 1 Pushbutton switch assemblies

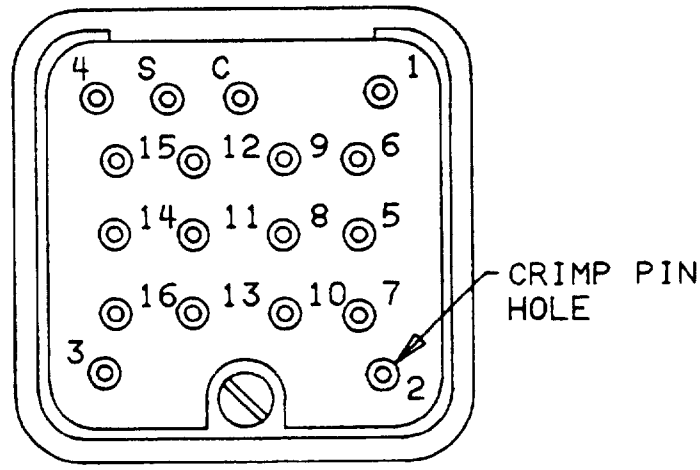
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STYLE S, W. & L



STYLE C

(COMMON TERMINATION SYSTEM)

FIGURE 1. Pushbutton switch assemblies - Continued.

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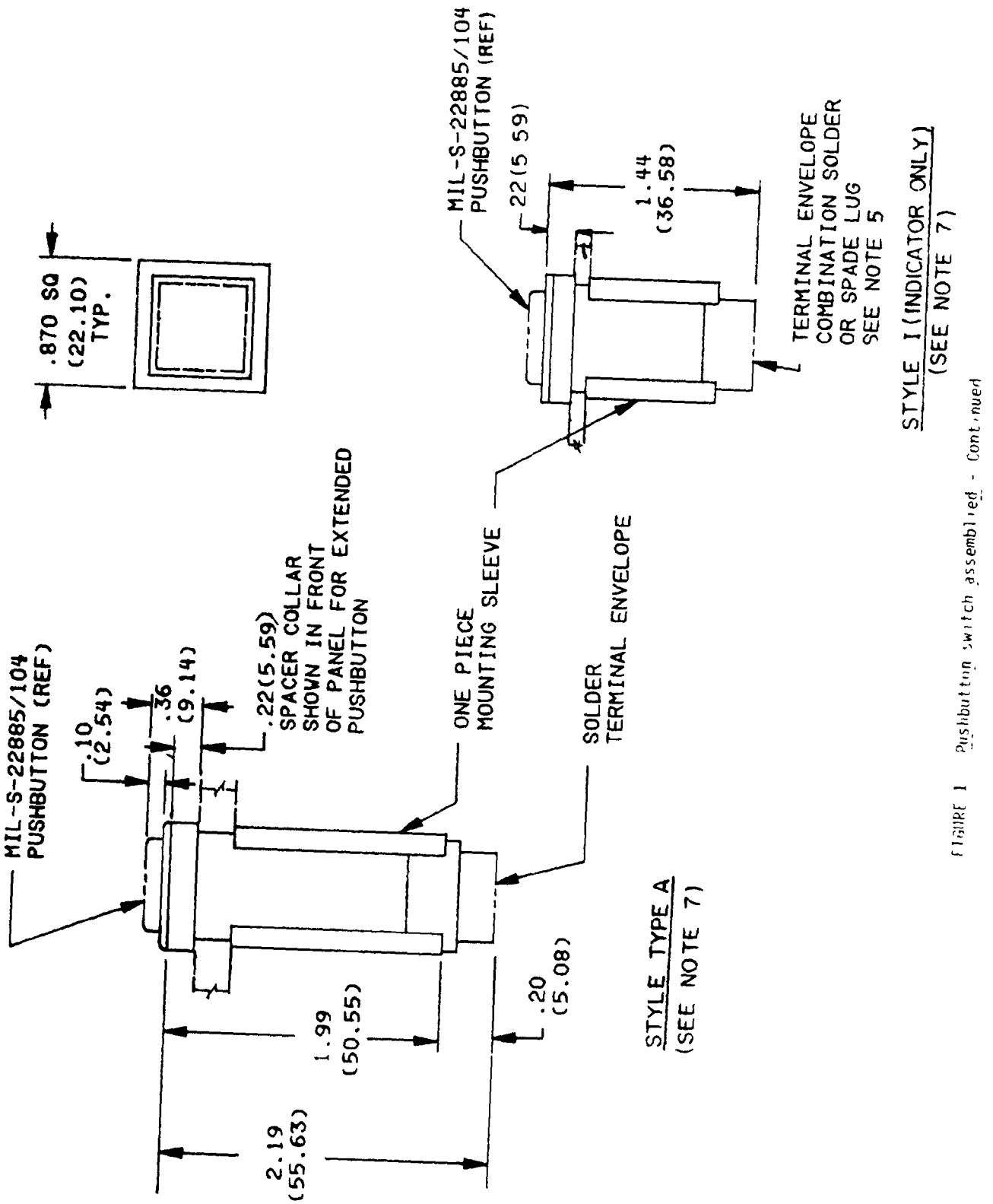
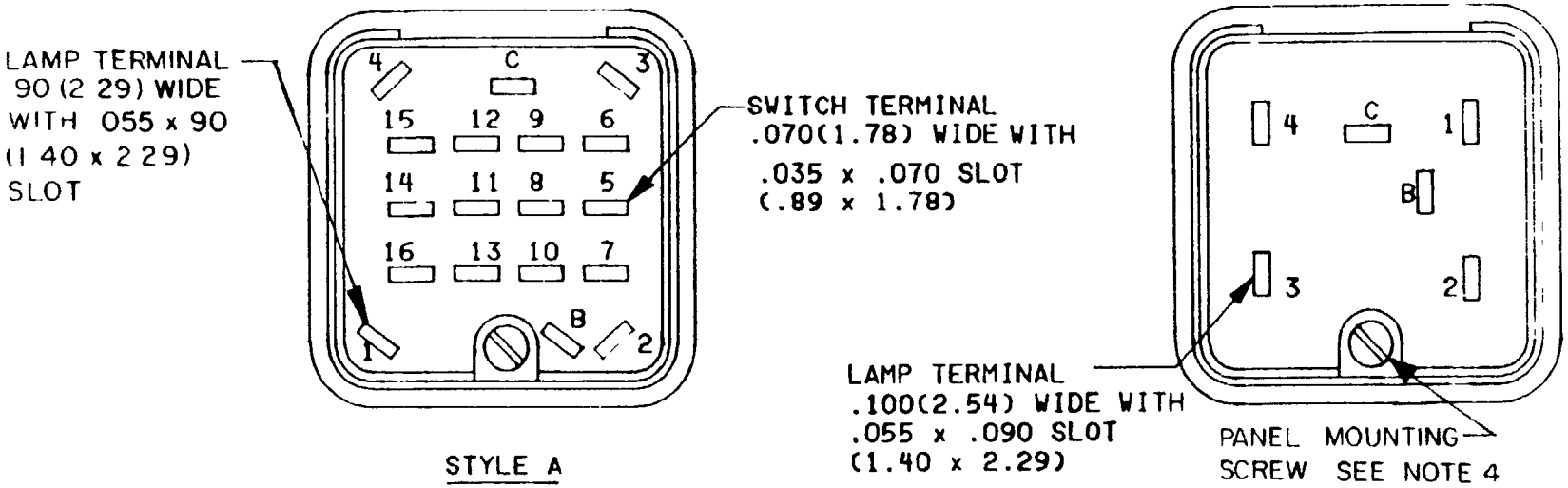


FIGURE 1. Pushbutton switch assembled - Continued

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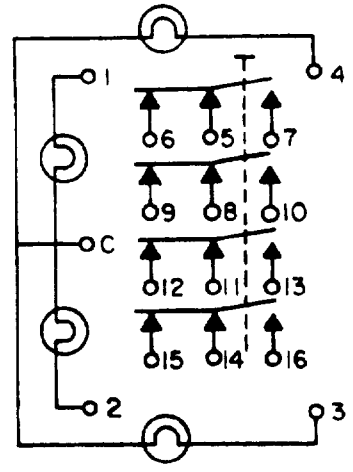


NOTES

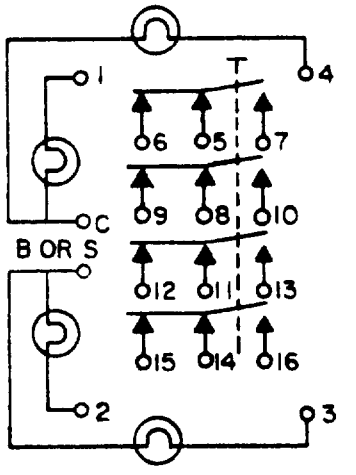
1. Dimensions are in inches. Metric equivalents are given for general information only.
2. Unless otherwise specified, tolerances are ± .015 (.38 mm) for three place decimals and ± .03 (.76 mm) for two place decimals.
3. Legend area for sunlight readable pushbutton display is .591 (15.01 mm) sq.
4. Maximum recommended tightening torque is 8 inch ounces on panel mounting screw.
5. Style I (indicator) terminals are a combination solder lug and spade lug. Each terminal will accept two 20 AWG stranded wires or a "fast-on" spade lug such as AMP #60537-1 or equivalent.
6. Unit is built with only the required terminals, see figure 2.
7. Style A and Style I are constructed with integral housing and switch/indicator, and do not accommodate MIL-S-22885/105 plug-in-modules.
8. Metric equivalents are in parentheses.

FIGURE 1. Pushbutton switch assemblies - Continued.

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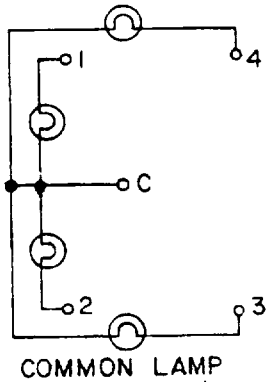


COMMON LAMP GROUND(CLG)

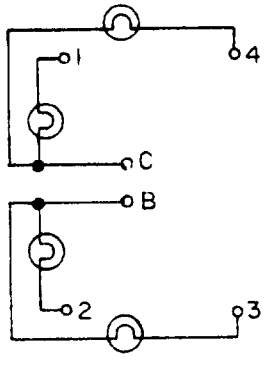


SPLIT LAMP GROUND (SLG)

SWITCH & LAMP SCHEMATICS, STYLE A, C, S, W & L



COMMON LAMP
GROUND (CLG)



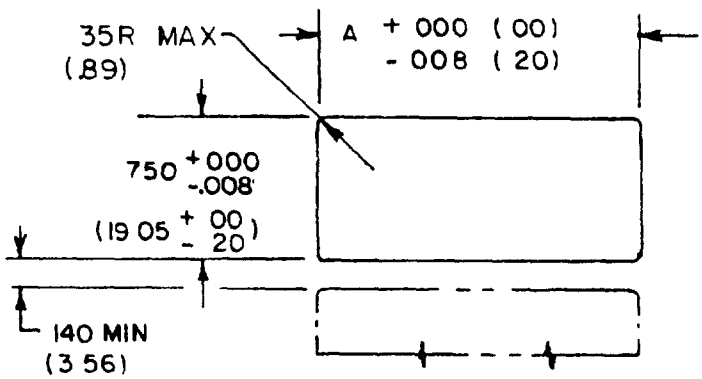
SPLIT LAMP
GROUND (SLG)

LAMP SCHEMATICS, STYLE I INDICATOR ONLY

NOTES
1 Delete pins 8 through 13 for 2PDT configurations
2 On momentary operation switches, terminals 6, 9, 12 and 15 are normally closed

FIGURE 2. Lamp and switch schematics

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No Units	Dimension A
1	750 (19 05)
2	1 621 (41.17)
3	2 492 (63 30)
4	3 363 (85 42)
N	A = 750 + 871 x N (19 05 + 22 12) x N

NOTES
1 Dimensions are in inches Metric equivalents are given for general information only
2 Unless otherwise specified, tolerances are ± 015 (38 mm) for three place decimals and ± 03 (76 mm) for two place decimals
3 Dimensions apply for both vertical and horizontal (shown) slot mounting
4 Metric equivalents are in parentheses

FIGURE 3 Horizontal and vertical mounting panel cutout dimensions

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REQUIREMENTS:

Design and construction See figures 1 and 2.

Complete switch shall consist of

1- Switch/Indicator housing Includes mounting sleeve, mounting screw with lockwasher, and spacer collar.

1- Replaceable switch/indicator module (styles S, W, L, and C only) in accordance with MIL-S-22885/105.

1- Pushbutton in accordance with MIL-S-22885/104. Accommodates 4 midget flanged base T-1 lamps. Pushbutton and lamps are acquired separately.

Classification

Enclosure design 1 (Unsealed).

Temperature characteristic 1(-55°C to +85°C).

Vibration grade. 3(10 - 2,000 Hz, 15 g peak).

Operation

A (Momentary action).

B (Alternate action).

H (Indicator only).

Color See MIL-S-22885/104.

Display type

S (Sunlight readable).

H (Hidden legend)

Material

Switch/indicator housing, spacer collar, and reversible mounting spacer Alluminum without EMI/RFI shielding Finished black in accordance with MIL-A-8625. With EMI/RFI shielding Chemical filmed in accordance with MIL-C-5541.

Mounting sleeve 300 series CRES in accordance with QQ-S-766, 2B finish.

Replaceable switch/indicator module See MIL-S-22885/105 (applicable to styles S, W, L, and C only).

Pushbutton See MIL-S-22885/104.

Terminal sealing grommet (style C switches) Silicone rubber

Common termination system (applicable to style C switches)

Terminal junction system Shall accept M39029/1-101 connector contacts. Shall accept MS27488-20 end seal plugs

Terminal sealing grommet Shall seal on smooth wire insulations of .138 (3.37 mm) to .083 (2.11 mm) diameter

Installation/removal tool M81969/14-02.

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Weight (±10%)

Style	Ounces	Grams
I	.71	20
A	1.20	34
C	1.90	54
S, W, and L	1.90	54

Operating characteristics

Actuation force 3.5 ±1.0 pounds (1.59 ± 45 kg)
Release force 0.4 (.18 kg) pound minimum.
Actuator travel .20 inch (5.08 mm) nominal.

Terminal strength

Common termination system (style C) Using crimping tool M22520/1-01 and positioner M22520/1-02 to install connector contact on wire, and insertion/extraction tool M81969/14-02, the crimped wire contact shall be inserted and extracted 10 times. The insertion and extraction force shall not exceed 10 pounds The contact shall again be inserted, and must hold axial load of 20 pounds minimum for 10 seconds.

Shock Method I.

Electrical endurance 25,000 cycles. See table I for test loads.

TABLE I. Electrical ratings.

Load	Sea level and 70,000 feet		Sea level
	28 V dc	115 V ac	30 mV 1/
Resistive	2 Amperes	2 Amperes	.01 Ampere
Inductive	1 Amperes	1 Amperes	
Lamp	.75 Ampere	.75 Ampere	

1/ Maximum dc or peak ac, 50,000 cycles. Low level life test conditions only.

Dielectric withstanding voltage

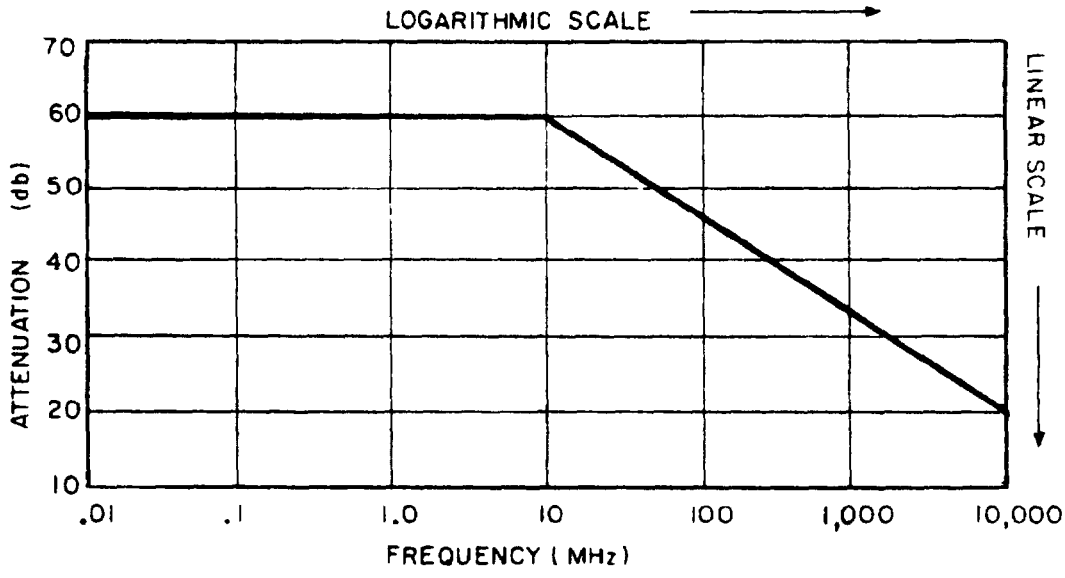
Sea level 1,056 V rms
70,000 ft 450 V rms.

Low level life Applicable.

EMI/RFI shielding Positive grounding and shielding efficiency test applicable when specified, see table III. The minimum shielding attenuation shall be as follows

Electric field	Minimum attenuation	Plane wave	Minimum attenuation
15 kHz	60 dB	100 MHz	47 dB
150 kHz	60 dB	400 MHz	39 dB
500 kHz	60 dB	1 GHz	33 dB
1 MHz	60 dB	2.5 GHz	28 dB
10 MHz	60 dB	10 GHz	20 dB

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Illuminated color See MIL-S-22885/104

Luminance See MIL-S-22885/104.

Sunlight readability Applicable, except units are tested for $\phi_1 = 45^\circ$ and $\phi_2 = 0^\circ$ only. See MIL-S-22885/104 for contrast ratios.

Marking. Part numbers for complete assemblies shall not appear on the items. The part numbers for individual plug-in modules and housing shall be marked on the items as required.

Terminal identification Circuit schematic shall be marked on the plug-in module. Terminal identification shall be shown on the housing terminal header or grommet.

Part number: M22885/103- (dash number, see table III). For acquisition of government spares for styles C, S, L and W, housing and switch module (per MIL-S-22885/105) shall be acquired separately.

Qualification inspection.

Group submission. Table II applies provided that similar products are qualified to MIL-S-22885/56 and MIL-S-22885/85.

Quality conformance inspection

Group B inspection

EMI/RFI shielding Only positive grounding test need be performed.

Group A inspection. Do not exceed 6 V dc or peak ac, 10 milliamperes during group A inspection.

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TABLE II. Qualification inspection (group inspection).

Qualification inspection table of MIL-S-22885			Additional inspections	Extent of approval
Switch	Group	Number of samples		
M22885/103-044 3/ 5/	I	All samples	7/ 8/	All
	II	4		
	III 1/	2		
	VI 2/	2		
	VII	2		
M22885/103-031 4/ 5/	I	All samples		
	II 1/	4		
	VIII	2 each color 6/		

1/ These inspections may be omitted, provided that similar products are qualified to MIL-S-22885/95.
2/ Resistive dc loads only (sea level and altitude).
3/ Units shall be equipped with M22885/104-10 pushbuttons and 4 MS24515-718 lamps.
4/ Units shall be equipped with M22885/104-7 pushbuttons and 4 MS24515-718 lamps.
5/ Legend shall consist of 2 lines of characters which utilize at least 3/4 of the maximum legend area.
6/ Two pushbuttons of each color may be installed on housings submitted to the inspections of group II.
7/ Two sample units each of styles S, W, L, A and I assemblies shall be inspected for visual and mechanical examination and operating characteristics.
8/ Two sample units each of style I indicator assemblies shall be inspected for solderability, resistance to soldering heat, and lamp contact resistance.

TABLE III Part numbers.

Part number M22885/103-	Style	Termination	Operation	Switch circuit	Lamp circuit	EMI/ RFI shielding
001	I	Solder	Indicator	N/A	CLG	No
002	"	"	"	"	SLG	No
003	"	"	"	"	CLG	Yes
004	"	"	"	"	SLG	Yes
005	A	"	"	"	CLG	No
006	"	"	"	"	SLG	No
007	"	"	"	"	CLG	Yes
008	"	"	"	"	SLG	Yes
009	"	"	Momentary	4PDT	CLG	No
010	"	"	"	4PDT	SLG	"
011	"	"	"	2PDT	CLG	"
012	"	"	"	2PDT	SLG	"
013	"	"	Alternate	4PDT	CLG	"
014	"	"	"	4PDT	SLG	"
015	"	"	"	2PDT	CLG	"
016	"	"	"	2PDT	SLG	"
017	"	"	Momentary	4PDT	CLG	Yes
018	"	"	"	4PDT	SLG	"
019	"	"	"	2PDT	CLG	"
020	"	"	"	2PDT	SLG	"
021	"	"	Alternate	4PDT	CLG	"
022	"	"	"	4PDT	SLG	"
023	"	"	"	2PDT	CLG	"
024	"	"	"	2PDT	SLG	"
025	F	Crimp pin	No module	4PDT	SLG	No
026	"	"	No module	4PDT	SLG	Yes
027	"	"	Indicator	N/A	CLG	No
028	"	"	"	"	SLG	No
029	"	"	"	"	CLG	Yes

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TABLE III. Part numbers - Continued.

Part number M22885/103-	Style	Termination	Operation	Switch circuit	Lamp circuit	EMI/ RFI shielding
030	C	Crimp pin	Indicator	N/A	SLG	Yes
031	"	"	Momentary	4PDT	CLG	No
032	"	"	"	4PDT	SLG	"
033	"	"	"	2PDT	CLG	"
034	"	"	"	2PDT	SLG	"
035	"	"	Alternate	4PDT	CLG	"
036	"	"	"	4PDT	SLG	"
037	"	"	"	2PDT	CLG	"
038	"	"	"	2PDT	SLG	"
039	"	"	Momentary	4PDT	CLG	Yes
040	"	"	"	4PDT	SLG	"
041	"	"	"	2PDT	CLG	"
042	"	"	"	2PDT	SLG	"
043	"	"	Alternate	4PDT	CLG	"
044	"	"	"	4PDT	SLG	"
045	"	"	"	2PDT	CLG	"
046	"	"	"	2PDT	SLG	"
047	S	Solder	No module	4PDT	SLG	No
048	"	"	No module	4PDT	SLG	Yes
049	"	"	Indicator	N/A	CLG	No
050	"	"	"	"	SLG	No
051	"	"	"	"	CLG	Yes
052	"	"	"	"	SLG	Yes
053	"	"	Momentary	4PDT	CLG	No
054	"	"	"	4PDT	SLG	"
055	"	"	"	2PDT	CLG	"
056	"	"	"	2PDT	SLG	"
057	"	"	Alternate	4PDT	CLG	"
058	"	"	"	4PDT	SLG	"
059	"	"	"	2PDT	CLG	"
060	"	"	"	2PDT	SLG	"
061	"	Solder	Momentary	4PDT	CLG	Yes
062	"	"	"	4PDT	SLG	"
063	"	"	"	2PDT	CLG	"
064	"	"	"	2PDT	SLG	"
065	"	"	Alternate	4PDT	CLG	"
066	"	"	"	4PDT	SLG	"
067	"	"	"	2PDT	CLG	"
068	"	"	"	2PDT	SLG	"
069	L	.40 wire wrap	No module	4PDT	SLG	No
070	"	"	No module	4PDT	SLG	Yes
071	"	"	Indicator	N/A	CLG	No
072	"	"	"	"	SLG	No
073	"	"	"	"	CLG	Yes
074	"	"	"	"	SLG	Yes
075	"	"	Momentary	4PDT	CLG	No
076	"	"	"	4PDT	SLG	"
077	"	"	"	2PDT	CLG	"
078	"	"	"	2PDT	SLG	"
079	"	"	Alternate	4PDT	CLG	"
080	"	"	"	4PDT	SLG	"
081	"	"	"	2PDT	CLG	"
082	"	"	"	2PDT	SLG	"
083	"	"	Momentary	4PDT	CLG	Yes
084	"	"	"	4PDT	SLG	"
085	"	"	"	2PDT	CLG	"
086	"	"	"	2PDT	SLG	"
087	"	"	Alternate	4PDT	CLG	"
088	"	"	"	4PDT	SLG	"
089	"	"	"	2PDT	CLG	"
090	"	"	"	2PDT	SLG	"
091	W	.73 wire wrap	No module	4PDT	SLG	No
092	"	"	No module	4PDT	SLG	Yes

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TABLE III. Part numbers - Continued

Part number M22885/103-	Style	Termination	Operation	Switch circuit	Lamp circuit	EMI/ RFI shielding
093	W	.73 wire wrap	Indicator	N/A	CLG	No
094	"	"	"	"	SLG	No
095	"	"	"	"	CLG	Yes
096	"	"	"	"	SLG	Yes
097	"	"	Momentary	4PDT	CLG	No
098	"	"	"	4PDT	SLG	"
099	"	"	"	2PDT	CLG	"
100	"	"	"	2PDT	SLG	"
101	"	"	Alternate	4PDT	CLG	"
102	"	"	"	4PDT	SLG	"
103	"	"	"	2PDT	CLG	"
104	"	"	"	2PDT	SLG	"
105	"	"	Momentary	4PDT	CLG	Yes
106	"	"	"	4PDT	SLG	"
107	"	"	"	2PDT	CLG	"
108	"	"	"	2PDT	SLG	"
109	"	"	Alternate	4PDT	CLG	"
110	"	"	"	4PDT	SLG	"
111	"	"	"	2PDT	CLG	"
112	"	"	"	2PDT	SLG	"

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(Project 5930-N574-1)

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